「Let F be a field of characteristic is <u>not</u> 2.
Let $D_1, D_2 \in F$, neither of which is a square in F.
If $\sqrt{D_1 D_2} \notin F$, then $F(\sqrt{D_1}, \sqrt{D_2})$ is degree 4 over F,
If $\sqrt{D_1 D_2} \in F$, then 〃 degree 2.

Proof. Clearly, $[F(\sqrt{D_1}) : F] = 2$, because $\sqrt{D_1} \notin F$ but $D_1 \in F$.
Suppose that $\sqrt{D_1 D_2} \notin F$. Then, $\sqrt{D_2} \notin F(\sqrt{D_1})$, because; if $\sqrt{D_2} \in F(\sqrt{D_1})$, then

$$\sqrt{D_2} = a\sqrt{D_1} + b \text{ for some } a, b \in F.$$

But, this implies

$$D_2 + a^2 D_1 - 2a\sqrt{D_1}\sqrt{D_2} = b^2 \Rightarrow \sqrt{D_1}\sqrt{D_2} = \frac{1}{2a}\cdot(b^2 - a^2 D_1 - D_2) \in F.$$

if $a \neq 0$. (Since $ch(F) \neq 2$, $2 \neq 0$.). And in case of $a = 0$,
$\sqrt{D_2} = b \in F$, so both case are contradiction
So, $[F(\sqrt{D_1}, \sqrt{D_2}) : F(\sqrt{D_1})] > 1$, exactly 2.

But, if $\sqrt{D_1 D_2} \in F$, then $\sqrt{D_1 D_2} = a$ for some nonzero $a \in F$. Now,

$$\sqrt{D_2} = \frac{a}{\sqrt{D_1}} \in F(\sqrt{D_1}),$$ so it has degree of 2.

More observation: let $a, b \in F$ be given, with $\sqrt{b} \notin F$. Then,

$$\sqrt{a + \sqrt{b}} = \sqrt{m} + \sqrt{n} \text{ for some } m, n \in F \text{ iff } \sqrt{a^2 - b} \in F.$$

Proof. Suppose the left assertion holds. Then,

$$a + \sqrt{b} = m + n + 2\sqrt{m}\sqrt{n} \Rightarrow a - (m+n) = 2\sqrt{m}\sqrt{n} - \sqrt{b} \in F.$$

So, $2\sqrt{mn} - \sqrt{b}$ must be zero, because if $2\sqrt{mn} - \sqrt{b} = k \neq 0$, then

$$2mn = k^2 + b + 2k\sqrt{b} \Rightarrow \sqrt{b} = \frac{1}{2k}(2mn - k^2 - b) \in F, \text{ contradiction.}$$

Thus, $2\sqrt{mn} = \sqrt{b}$ and $a = m + n$, $a^2 - b = (m+n)^2 - 4mn = (m-n)^2$,
or $\sqrt{a^2 - b} = m - n \in F$. Conversely, if $\sqrt{a^2 - b} \in F$, then take

$$m = \frac{1}{2}(a + \sqrt{a^2 - b}) \text{ and } n = \frac{1}{2}(a - \sqrt{a^2 - b}). \text{ Then,}$$

$$(m-n)^2 = a^2 - b, \ 4mn = b, \text{ so } (\sqrt{m} + \sqrt{n})^2 = m + n + 2\sqrt{mn} = a + \sqrt{b}.$$

Using the observation above, for $a, b \in \mathbb{Q}$,

$\mathbb{Q}(\sqrt{a+\sqrt{b}})$ is biquadratic if and only if

$$\sqrt{a^2-b} \in \mathbb{Q} \quad \text{and} \quad \sqrt{\frac{a \pm \sqrt{a^2-b}}{2}} \notin \mathbb{Q}.$$

「Example」

Evaluate the degree of $\mathbb{Q}(\sqrt{3+2\sqrt{2}})$ over $\mathbb{Q}$

Since $\sqrt{3+2\sqrt{2}} = \sqrt{3+\sqrt{8}}$, $9-8=1$ is square in $\mathbb{Q}$.

But, $\sqrt{\frac{3 \pm \sqrt{1}}{2}} = 1$ or 0, squares, it is not biquadratic.

Specifically, take $m = \frac{1}{2}(3+1) = 2$, $n = \frac{1}{2}(3-1)$, so

$\sqrt{3+2\sqrt{2}} = \sqrt{m} + \sqrt{n} = \sqrt{2}+1$, so $\mathbb{Q}(\sqrt{3+2\sqrt{2}}) = \mathbb{Q}(\sqrt{2}+1) = \mathbb{Q}(\sqrt{2})$.

Degree 2.

Similarly, for $\sqrt{3 \pm 4i}$, take $m = \frac{1}{2}(3+5) = 4$, $n = \frac{1}{2}(3-5) = -1$.

$\sqrt{3+4i} = 2+i$, and $\sqrt{3-4i} = 2-i$. so,

$\mathbb{Q}(\sqrt{3+4i} + \sqrt{3-4i}) = \mathbb{Q}$.

Galois Group of Biquadratic Extension.

Let F be a field, and $D_1, D_2 \in F$ s.t. $\sqrt{D_1}, \sqrt{D_2}, \sqrt{D_1 D_2} \notin F$.

Then, $K = F(\sqrt{D_1}, \sqrt{D_2})$ is biquadratic.

Moreover, $\mathrm{Gal}(K/F) \cong \mathbb{Z}_2 \times \mathbb{Z}_2$.